

(Please write your Exam Roll No. immediately)

Exam Roll No: _____

END TERM EXAMINATION
THIRD SEMESTER [B.TECH.] DECEMBER 2025 - JANUARY 2026
Paper Code: CIC-205 • Subject: Discrete Mathematics

Time: 3 Hours

Maximum Marks: 60

Note: Attempt all questions as directed. Internal choice is indicated. **Q. No. 1** is compulsory (attempt any 5 parts).

Q. No.	Question	Max. Marks	CO(s)
Q1	Attempt any FIVE of the following questions:	[4 × 5 = 20]	CO1-CO4
(a)	Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$. (i) Is R reflexive? (ii) Is R symmetric? (iii) Is R transitive?	[4]	CO2
(b)	What is the principle of inclusion-exclusion? Provide the mathematical formula for two finite sets $ A \cup B $.	[4]	CO1
(c)	Define generating functions and give an example. List two real-world computer science applications of divide-and-conquer algorithms.	[4]	CO2
(d)	What is Master's theorem? State its importance in algorithm time complexity analysis.	[4]	CO2
(e)	Prove that the intersection of two subgroups $H_1 \cap H_2$ of a group G is also a subgroup of G.	[4]	CO3
(f)	Explain Lagrange's theorem for finite groups and its direct corollaries.	[4]	CO3
(g)	Prove that for any connected planar graph, $V - E + F = 2$, where V is the number of vertices, E the number of edges, and F the number of faces/regions.	[4]	CO4
(h)	What is a spanning tree? Give an example of a connected graph with 5 vertices and find its minimum spanning tree (MST).	[4]	CO4
SECTION A		[10]	CO1/CO2
Q2	Using the Pigeonhole Principle, prove that in any group of 13 people, at least two people were born in the same month. Also prove by contradiction that $\sqrt{2}$ is irrational.	[10]	CO1
OR			
Q3	Define an equivalence relation R on the set of integers $\mathbb{Z}$ as: $a \sim b$ if and only if $a - b$ is divisible by 3. (a) Verify that R is an equivalence relation (reflexive, symmetric, transitive). (b) Find the equivalence classes [0], [1], and [2].	[10]	CO2
SECTION B		[10]	CO2/CO3
Q4	(a) Convert the following formula to Conjunctive Normal Form (CNF): $(P \rightarrow Q) \wedge (\neg R \rightarrow P)$. (b) Define a lattice. Show that the set $\{2, 4, 8, 16\}$ with divisibility as the partial order forms a lattice. (c) Minimize the following Boolean expression using a 3-variable Karnaugh map: $F(A, B, C) = \sum m(0, 1, 2, 5, 7)$.	[10]	CO3
OR			
Q5	Analyze the time complexity of Merge Sort using recurrence relations and solve it. Also find the asymptotic time complexity of the following recurrence relation using Master's Theorem: $T(n) = 4T(n/2) + n^2$	[10]	CO2

Q. No.	Question	Max. Marks	CO(s)
SECTION C		[10]	CO3
Q6	Define a group homomorphism. Prove that the image $f(G)$ of a group homomorphism $f : G \rightarrow G'$ is a subgroup of the codomain G' . In any group G , prove that the identity element is unique and the inverse of each element is unique.	[10]	CO3
OR			
Q7	Let $G = \{e, a, b\}$ be a group of order 3 under multiplication. Use Cayley's theorem to construct the left regular representation of G and show that G is isomorphic to a subgroup of the symmetric group S_3 .	[10]	CO3
SECTION D		[10]	CO4
Q8	Analyze and compare the time complexities of Prim's algorithm and Kruskal's algorithm for finding Minimum Spanning Trees. Which algorithm would you prefer for dense graphs vs. sparse graphs? Justify your answer. Also state and prove the Five Color Theorem for planar graphs.	[10]	CO4
OR			
Q9 (a)	Write an algorithm (using BFS or DFS) to find all connected components of an undirected graph. Apply your algorithm to the graph: $V = \{1, 2, 3, 4, 5, 6\}$ and edge set $E = \{(1, 2), (2, 3), (4, 5)\}$. List the connected components found.	[5]	CO4
Q9 (b)	Does the following graph contain a Hamiltonian circuit ? Justify your answer with a formal argument or cycle decomposition: $V = \{1, 2, 3, 4\}$ $E = \{(1, 2), (2, 3), (3, 4), (4, 1), (1, 3)\}$	[5]	CO4